

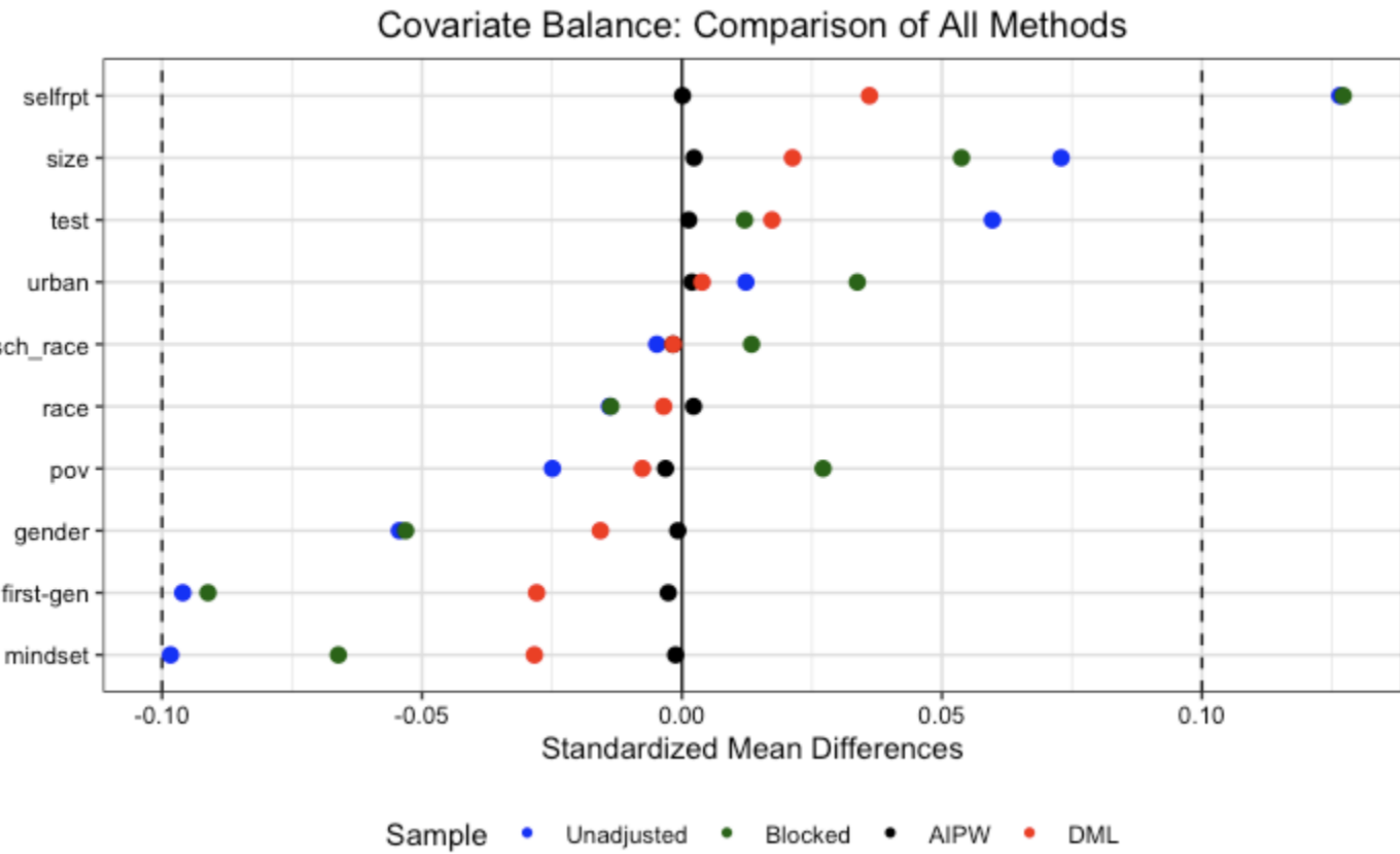
Introduction

The goal of this analysis is to investigate the causal effect of intervention designed to instill a growth mindset in students. In this analysis, I try to find whether the intervention has meaningful impact or not.

First, I perform an in-depth exploratory data analysis on the dataset to identify confounding covariates. Based on my learnings, I applied multiple causal inference methods to estimate the Average Treatment Effect (ATE). I employed the following methods : Unadjusted Estimate, Blocked Design, AIPW (Augmented Inverse Probability Weighting) and DML (Double Machine Learning).

My whole analysis revolves around the following hypothesis test at significance level of 0.05. Null Hypothesis (H_0) : The treatment effect is zero. Alternate Hypothesis (H_a) : The treatment effect is not zero.

By above mentioned methods, I evaluated whether the intervention produced has statistically significant effect in growth mindset. The following sections will go into details.



Dataset Description & Exploration

The data is a synthetically generated, so I treated this analysis as an observational study. The dataset consists of 10 covariates (four student-level covariates and six school-level covariates), the outcome Y and treatment Z. The dataset also contains school ID, which I will be not be using in my analysis.

The dataset has 10391 rows and 13 columns, it has no null values and it has 4 categorical covariates (race, gender, urban, fgen (binary)) and 6 continuous covariates (selfrpt, mindset, test, sch_race, pov, size).

For Exploration, I generated love-plots for different covariates with standardized mean differences. In the figure, blue dots indicate sample data. Love plot shows the difference in covariate values between treatment and control groups. The mean shifts are significant, instead of being very close to zero like in a completely random experiment. Note selfrpt and mindset, which are measured prior to the experiment have the most extreme mean differences.

This shows that confounding is present in the data. The treatment assignment is not independent of the covariates.

Methodology

The goal is to estimate the treatment effect as accurately as possible. The Average Treatment Effect (ATE) is the average difference in outcomes if everyone got treatment vs. if no one did. [1]. For my analysis, I took a step-by-step approach, starting from the simplest to the semi-parametric machine learning approach.

1. Simple Additive Linear Regression (Unadjusted estimate)

I ran a basic linear regression, essentially on the raw difference in average scores between students who got the nudge and those who didn't, this can be seen in equation below.

$$\hat{\tau} = \bar{Y}^{obs}(1) - \bar{Y}^{obs}(0)$$

This method assumes no interference (SUTVA), strong ignorability on pretreatment covariates, and overlap. This doesn't handle covariant imbalance. That makes this method a useful baseline, but a biased one. Recognizing the limitations, I employed the blocked design method.

2. Blocked Design : The goal was to make homogenous groups of units based on some secondary source of variation among units, covariates [1]. Then randomize treatments within blocks separately. In doing so, we will guarantee balance (between treatment and control groups) on whichever covariates we use to do the blocking. In this method, we treat each block as a random distribution and calculate the estimates just like in first method, then we combine all of them by doing weighted average.

This method, in addition to SUTVA and pretreatment covariates, assumes treatment is ignorable within each block and that both treated and control units exist in every block. After experimenting with various covariates, I stratified data into blocks based on test scores (test) and poverty levels (pov), to create a more comparable treatment and control groups. This reduces covariate imbalance compared to our first method. But, the love plot from the figure shows that even with Blocked design there is still imbalance.

3. AIPW

To address the biases, I used Augmented Inverse-Probability Weighting. In this method, we first model who gets the nudge (the propensity score) and then fit separate outcome models for treated and untreated students as seen in equation below, then I trimmed extreme scores to keep the weights stable. AIPW lets us combine weighting and regression so that, even if one model was incorrect, the model would still work. This is called double robustness. This method, requires SUTVA, ignorability and overlap on pretreatment covariates. From the love plot, we can see that this significantly reduces the covariate imbalance. This method was re-using the same data twice in the model which leads to double-dipping. We can get around this by using a separate sample to estimate [1] but this leads to loss in precision.

$$\tilde{\mu}_{1,dr} = \frac{1}{n} \sum_{i=1}^n \left(\underbrace{\frac{W_i Y_i}{\pi}}_{\text{IPW}} + \underbrace{\tilde{\mu}_1(\mathbf{x}_i) \left(1 - \frac{W_i(\mathbf{x}_i)}{\pi} \right)}_{\text{augmentation}} \right)$$

So, I decided to try semi-parametric ML model called Double Machine Learning which is more flexible. This allows me to use full dataset without running into issues like double dipping.

4. DML

Finally, I used Double Machine Learning, which splits the data into folds, trains the nuisance models on one part, and estimates effects on the hold-out part. There is no problem of overfitting or bias from re-using data [2]. Though from the love plot we can see that this doesn't perform as well as the AIPW.

For all the above methods, I used bootstrapping with 1,000 replications, this leads to a more reliable causal inference, as this accounts for variability in the observational study.

Results

- Our Hypothesis test rejects the null hypothesis for all methods, this shows that the results are statistically significant [table].
- We can see that, the standard error decreases as we go from simple to more complex models.
- Unadjusted simple additive model gave the worst performance of all as it doesn't consider covariate imbalance.
- Whereas, Blocked Design is a clear improvement [fig, table]. It's confidence interval is narrower and the standard error decreased, It handled covariant imbalance but not quite well.
- AIPW performs best in covariate balance, Figure shows mean differences for covariates near zero line , showing it handled covariate imbalance the best.
- Why AIPW outperforms DML:** AIPW corrects for both treatment and outcome models on the full dataset, thus giving better covariate balance despite of training on same data twice (double-dipping), whereas DML didn't do as good of a job as AIPW as most of the heavy lifting is being done by the hold-out part.
- An ATE of + 0.41 means that, on average, students who received the intervention scored 0.41 points higher on the outcome measure compared to if they hadn't received the intervention.

Method	ATE Estimate	Std Error	95% CI	Statistical significance
Unadjusted	0.4569	0.0144	[0.4287,0.4851]	Yes
Blocked Design	0.4729	0.0143	[0.4448,0.5009]	Yes
AIPW (double - dipping)	0.4118	0.0143	[0.3831,0.4407]	Yes
DML	0.4175	0.0123	[0.3934,0.4416]	Yes

Conclusion

The analysis shows a statistically significant treatment effect across all methods, we can conclude that the intervention does have significant effect on growth mindset in students. Each successive method improved in handling covariance imbalance and gave less biased results. AIPW with double dipping handled imbalance the best, showing that additional complexity in model is unnecessary. I demonstrated how starting from simple linear regression we can move up the complexity as we address different shortfalls.

References

- Lecture Notes
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2024). Double/Debiased Machine Learning for Treatment and Structural Parameters. arXiv preprint arXiv:1608.00060.